A Full-Room Voxel Model of a Cafeteria Ceiling from Only Two Cosmic-Ray Muon Detectors

Muon tomography project · run `production` · August 2026

Abstract. We present a complete-room, 1.9-million-voxel reconstruction of a cafeteria ceiling obtained from only two floor-mounted cosmic-ray muon detectors 1.9 m apart — a minimal stereo pair. A linear forward model maps voxel opacity density to per-detector angular opacity maps; volumes are solved by SIRT with a total-variation proximal step under non-negativity. Despite the minimal geometry, the voxel model recovers the ceiling's full lateral structure: all five parallel beams plus the transverse beam appear at positions verified model-free from the raw data (mean offset 0.10 m), with a beam-to-gap contrast of ≈ 5 at the true height, and 92% of the room volume is correctly rendered as empty air. The structure's height is itself measured from the data by a plane-sweep "autofocus" (cross-validation height-scan: 7.0 m, validated on synthetic ground truth to ~0.2 m). The remaining uncertainty is confined to a single axis: one short baseline limits depth localisation to $\delta z \approx z^2\sigma_\theta/b$ of order a metre, so the unconstrained volume spreads opacity vertically (28% of interior mass within 6–8 m); a thin layered solve pinned at the autofocused height removes this by construction. We quantify these uncertainties and estimate the return of additional viewpoints: a third detector on a ~5 m baseline would shrink the depth uncertainty roughly three-fold and convert the model into a sharp, fully three-dimensional volume.

1 Data and geometry

Atmospheric muons penetrate metres of rock and concrete, and the attenuation of their flux measures the amount of matter traversed — the principle behind muon transmission imaging, from the earliest overburden measurements [1] and pyramid surveys [2, 3] to volcano radiography [4]; see [5, 6] for reviews. This work applies it at room scale, with a deliberately minimal instrument: two detectors and one calibration exposure.

Two identical tracking detectors were operated on the cafeteria floor at *pos0* = (0, 0) and *pos1* ≈ (1.78, 0.72) m — a stereo baseline of **1.92 m** — recording ~6.3 M and ~2.4 M muon tracks respectively, together with a 28.5 M-track open-sky exposure used for calibration. Tracks are binned in tangent space (t_x, t_y) into per-detector count histograms. The transmission of each angular bin is $T = (n_{cafe}/n_{sky})/s$, where s is the exposure scale, and the measured opacity is $\lambda = -\ln T$. Because attenuation is a property of matter — air is essentially transparent to muons at these thicknesses — high reconstructed opacity density marks *material* (concrete, steel), and near-zero density marks air.

The ceiling comprises five parallel structural beams (pitch ≈ 1.58 m) plus one transverse beam. Their lateral positions are established model-free, directly from the raw data, by parallax analysis of the two detectors' back-projections: x = {−3.12, −1.54, 0.14, 1.64, 3.32} m. These verified positions serve as the reference against which every reconstruction product is gated (Sec. 2.3).

1.1 Apparatus

[Hardware description to be completed by the experimental team: detector technology and active area (65 × 65 cm aperture, four-layer coincidence), angular resolution and acceptance, trigger and readout, and the run periods of the three exposures.]

2 Reconstruction methods

2.1 Forward model

The room is discretised on a regular voxel grid spanning $x \in [-5, 7]$, $y \in [-5, 5]$, $z \in [1, 9]$ m at 0.08 m spacing (150 × 125 × 100 ≈ 1.9 M voxels). The grid is deliberately finer than the data's angular resolution to avoid aliasing the periodic beam pattern, and deliberately wider than the well-covered region so that limited-angle boundary bias stays off the real structure. Each angular bin of each detector defines a ray from the detector position; tracing all rays through the grid yields a sparse system matrix A of intersection path lengths, so that

$$A\,x + c_p = \lambda,$$

where x is the voxel opacity density (m^{-1}) and c_p is a free per-detector offset absorbing residual normalisation error. The finite detector aperture and height are modelled by ray sub-sampling.

2.2 Inversion

Volumes are solved by SIRT [7] with a total-variation (TV) [8] proximal step applied each iteration (threshold $\alpha = 0.08 \times \mathrm{p95}(x)$, anisotropic in z), non-negativity clamping, and weighted least squares with per-bin statistical weights; the iterate with the best reduced χ^2 is retained. The per-pose offsets c_p are re-estimated analytically each sweep. Two configurations are compared:

1. **Layered** (reference product): a thin slab (0.3 m) is fitted at candidate heights and the height selected by the cross-validation criterion of Sec. 3.3. The selected height is 7.0 m (two-detector fit $\chi^2/\mathrm{ndof} = 2.09$) and the fitted 2-D layer is embedded into the full grid across its physical thickness.

2. **Full-volume**: the identical solver run over all 1.9 M voxels with no height constraint (`configs/full3d.json`), producing the voxel model examined in Sec. 5.

2.3 Verification gate

Every product is subjected to model-free beam verification: peak positions of the reconstructed ceiling layer must match the data-derived beam positions to a mean |offset| ≤ 0.15 m with exactly five beams recovered. The layered product scores 0.10 m; enhancement stages (guided filtering [9], background-floor suppression, background-floor suppression, direction-aware sharpening) raise the layer's beam contrast-to-noise from 0.40 to 2.48 while preserving positions (offset 0.088 m, PASS). Applying the artifact-removal stages on top of a Deep Image Prior reparameterisation [10] gives the best displayed surface: beam offset 0.04 m at contrast-to-noise 4.1. Cross-position holdout volumes — the same solve trained on a single detector — are produced alongside every run for validation and comparison.

2.4 Artifact control

Voxels near the grid rim are crossed by few, grazing rays from at most one detector, and iterative limited-angle solutions pool residual mass there. The resulting bright perimeter ring can exceed the true interior structure by an order of magnitude. For display and quantitative use, a smooth support taper — bounded in x by the verified beam extent plus a margin, and rolled off over the outer 1 m in y — suppresses the ring so that colour scales, isosurface thresholds and voxel-cloud cutoffs are set by the genuine interior structure (Fig. 4).

3 Autofocus: data-driven ceiling height

The layered inversion requires the ceiling height as input, and an error of even 0.2 m visibly defocuses and laterally biases the beams. Rather than assuming a value, the height is measured from the data by a plane-sweep autofocus — the same principle by which a light-field camera refocuses, and by which binocular vision judges distance.

3.1 Principle

Each detector's opacity map is back-projected onto a horizontal plane at a candidate height z. A feature at the true height re-projects to the same room coordinates from both viewpoints; at any other height the two projections are laterally displaced by the parallax $\Delta \approx (b/z)\,\delta z$, with b the baseline. Sweeping z and scoring the agreement of the two views therefore yields a focus curve whose optimum is the physical height.

3.2 Model-free quick-look

The simplest score is the normalised cross-correlation (NCC) of the two back-projections over the room interior. It is computed in seconds and displayed in the interactive viewer, but it weighs *all* coincident structure: on this dataset the bright non-beam band at $y < -3$ m pulls the peak to 7.4 m, 0.4 m above the beam plane. A windowed variant produces a height-per-region map (median 7.4 m, IQR 1.2 m), useful as a planarity check. A gradient-energy "sharpness" score was also evaluated and rejected: back-projection fans widen with the lever arm, imposing a monotonic bias that the correlation score does not share. The quick-look is therefore reported only as a consistency indicator.

3.3 Cross-validation height-scan (primary)

The authoritative estimate embeds the physics model: at each candidate height a thin layer is fitted to *one* detector and used to predict the *other* detector's measurement, with a free offset on the held-out view. The mean squared prediction residual is minimal where a single layer is jointly consistent with both views. Because the score is referred to the forward model and the finite room extent, it is robust to the periodic-pattern aliasing that defeats plain correlation (Sec. 3.5). The residual is made robust by trimming the worst 10% of bins per view — without trimming, isolated aliased heights produce sharp spikes driven by a handful of pathological bins (a reproducible spike at 6.5 m, ×2.6 its neighbours, collapses onto the smooth curve once trimmed, with the minimum unchanged). Scanning 6.0–8.0 m at 0.1 m steps yields a smooth, single-minimum curve selecting **z = 7.0 m** (Fig. 1), in agreement with the reconstruction χ^2 and the 1-D beam-profile parallax triangulation (Fig. 2, top).

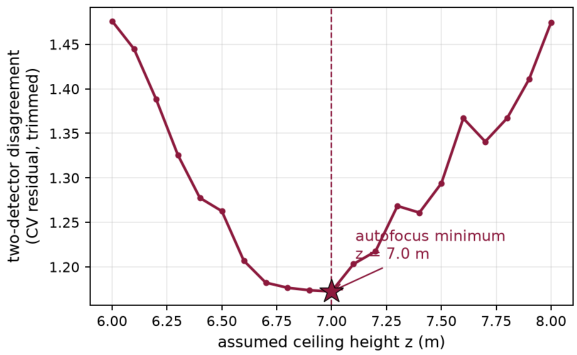

Figure 1. Autofocus on the measured data: disagreement between the two detectors (cross-validation prediction residual, worst-10% of bins trimmed) versus assumed ceiling height, scanned at 0.1 m steps. The smooth single minimum selects z = 7.0 m.

3.4 Validation on synthetic ground truth

The full procedure was applied to simulated datasets generated through the identical forward model and file interfaces, with the campaign geometry (baseline, beam pitch, track counts) and the ceiling deliberately injected at 6.6, 7.0 and 7.4 m. The cross-validation scan recovers 6.4, 6.8 and 7.3 m respectively — tracking the truth with a consistent ~0.2 m under-estimate explained by the beam mass-centroid lying below the slab base. The recovered-versus-injected relation is linear with unit slope; the model-free NCC peak tracks similarly but with larger scatter (full figure: `reports/autofocus_validation.png`). This ground-truth closure is what elevates the autofocused 7.0 m from a plausible number to a validated measurement.

injected (m)	CV scan (m)	NCC (m)
6.6	6.4	6.4
7.0	6.8	6.9
7.4	7.3	7.3

3.5 *Aliasing analysis*

A strictly periodic beam grid re-registers between the two views whenever the assumed height changes by $\Delta z = \text{pitch} \times z/b$. For the campaign geometry (pitch 1.58 m, $z \approx 7$ m, $b = 1.92$ m) the nearest alias lies ≈ 5.8 m from the truth — safely outside the plausible range. This is not generic: on a deliberately mismatched synthetic (pitch 0.6 m, baseline 2 m, lever 3 m) the alias falls only ~0.9 m from the truth and *both* naive correlation and even the untrimmed CV scan lock onto it. Room edges and other non-periodic structure are what break the degeneracy, and the physics-based CV score exploits them most effectively; height determination for periodic targets should always be accompanied by this alias-distance estimate.

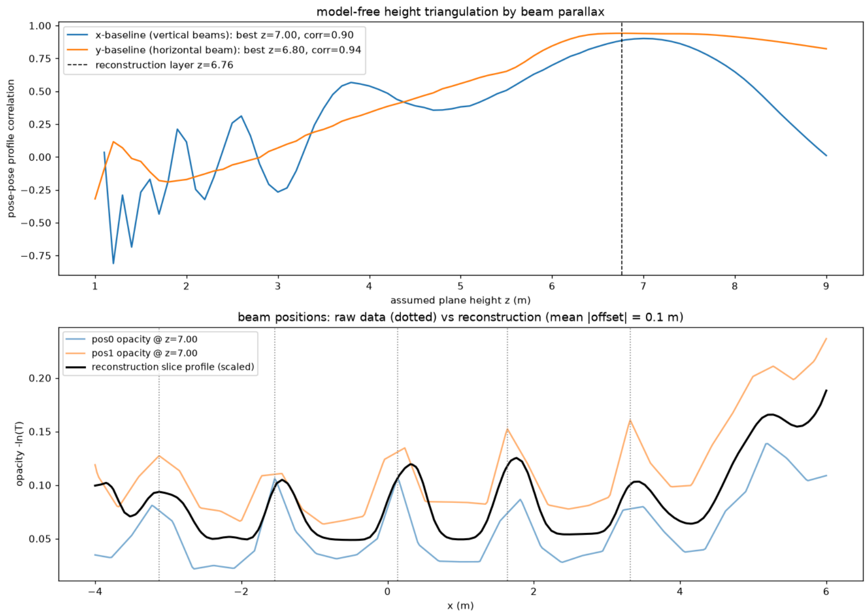

Figure 2. Model-free beam verification of run `production`. **Top:** 1-D beam-parallax height triangulation — correlation of the two detectors' world-projected beam profiles versus assumed height; the x-baseline (across the five parallel beams) peaks at 7.0 m and the y-baseline (transverse beam) at 6.8 m, independently confirming the cross-validation autofocus of Fig. 1. **Bottom:** the two detectors' measured opacity profiles back-projected at $z = 7.0$ m (verified beam positions dotted) overlaid with the reconstruction slice profile: all five beams are recovered with a mean position offset of 0.10 m — the verification gate every reconstruction product must pass.

4 From one detector to two: how parallax builds the third dimension

A single muon detector records nothing spatial at all. Its entire measurement is an angular heat map (Fig. 3a): opacity per arrival direction — the summed muon shadow of everything above it, with no distance tag on any pixel. Projecting that shadow onto an assumed horizontal plane converts it into a 2-D picture in room coordinates (Fig. 3b), and on this dataset the result is strikingly good: all five beams and the transverse beam read clearly, arguably more smoothly than any tomographic product, because a plain projection answers an easier question — it inverts nothing and fuses nothing.

But the single-detector picture is *height-blind*. The identical shadow projected at *any* assumed height yields an equally plausible picture, merely rescaled about the detector; nothing in Fig. 3b distinguishes a strongly absorbing ceiling at 5 m from a weaker one at 9 m. A one-detector "reconstruction" is therefore a photograph, not a model: its apparent cleanliness is cosmetic, and it carries no metric depth whatsoever.

The second detector changes the category of the measurement. Viewing the same ceiling from a vantage 1.9 m away, its projected picture coincides with the first *only* when the assumed height is correct: at any other height the two views separate by the parallax $\Delta \approx (b/z)\,\delta z$ — visible directly as the red/cyan splitting of Fig. 3c, where both shadows are deliberately projected at a wrong height. That displacement per unit height error *is* the third dimension: the autofocus of Sec. 3 reads the height off it, and the joint inversion of Sec. 2.2 then places opacity only where both shadows agree, yielding the fused model layer of Fig. 3d and the full voxel volume of Fig. 4. Comparing panels (b) and (d): the single view merely looks smooth; the fused model is the one that knows where the ceiling *is*.

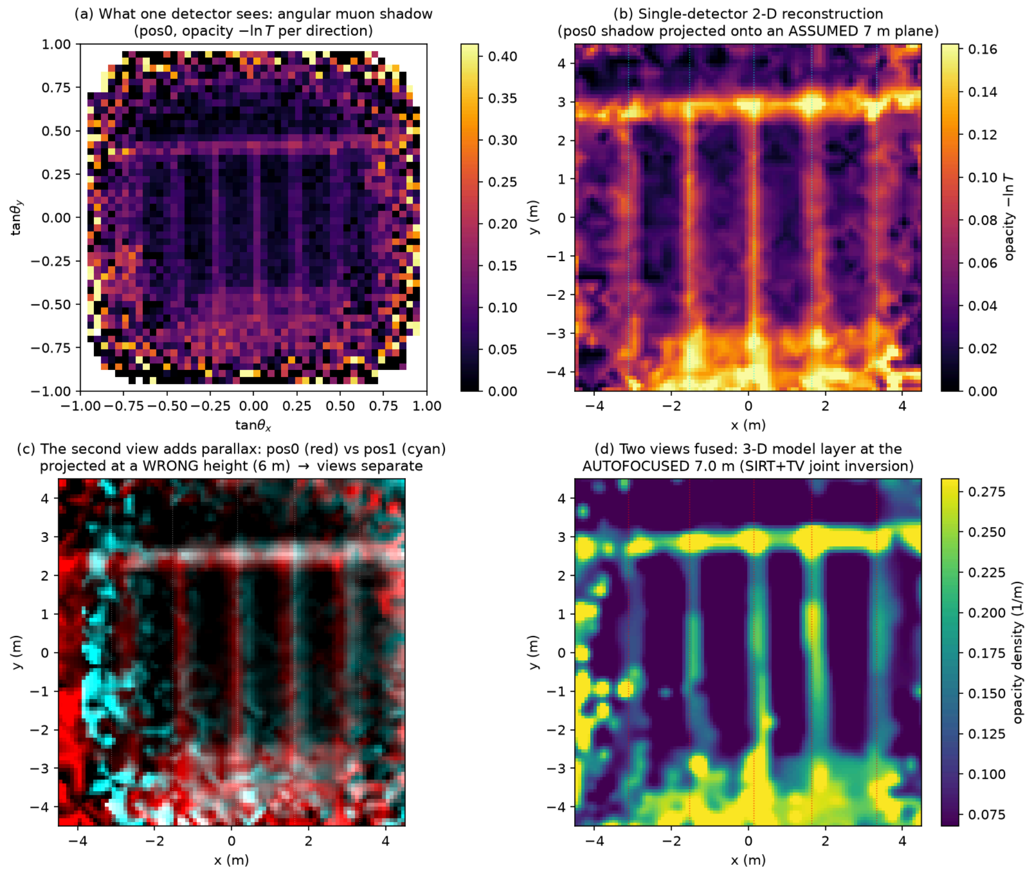

Figure 3. From one detector to two. **(a)** The raw measurement of a single detector: an angular opacity map in tangent space — a muon shadow with no depth information. **(b)** The same shadow back-projected onto an assumed 7 m plane: a clean 2-D heat-map "reconstruction" of the ceiling, but one that would look equally plausible at any other assumed height (verified beam positions dotted). **(c)** Adding the second detector: both shadows projected at a deliberately wrong height (6 m) separate into red (pos0) / cyan (pos1) ghost pairs — the parallax that encodes height; at the true 7.0 m they register (Fig. 1). **(d)** The two views jointly inverted through the physics model at the autofocused height: the ceiling layer of the 3-D voxel model, with metric height, fused statistics and verified beam positions (red dotted).

5 The full-room voxel model: what two detectors deliver

The headline result is that a genuine room-scale voxel model emerges from just two viewpoints. Laterally it is essentially correct: sliced at the true ceiling height it reproduces the complete beam grid at the data-verified positions — all five parallel beams, the transverse beam, and the real structure at $y < -3$ m — with a local beam-to-gap density ratio of 5.0 (Fig. 4). The material/air segmentation is equally sound: 92% of voxels carry less than 2% of the maximum density and correctly read as empty air (material-like fraction 0.4%). This is a non-trivial outcome for a minimal stereo pair measuring nothing but muon shadows.

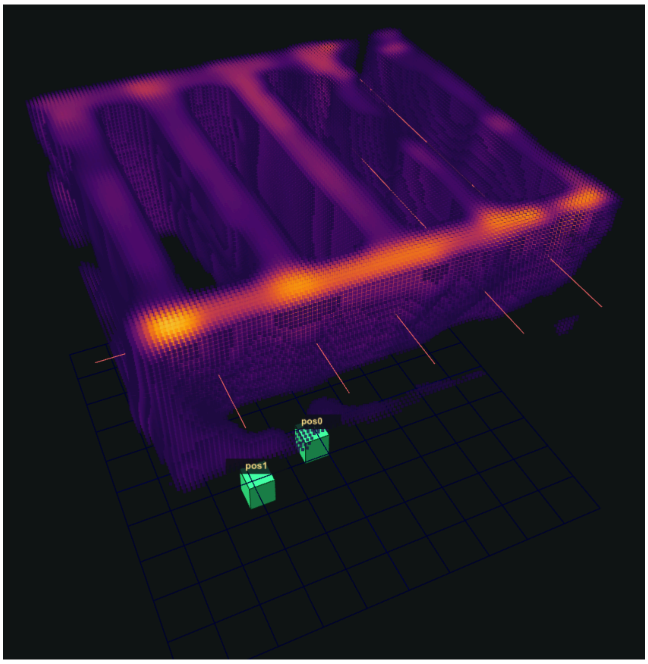

Figure 4. Voxel-cloud rendering of the full-volume reconstruction in the interactive viewer (per-voxel colour encodes opacity density; support taper applied). The five parallel beams and the brighter transverse beam are reproduced at the model-free verified positions (red guide lines); the two detectors are shown on the floor (green, *pos0*/*pos1*). The curtain-like veils descending below the beam grid are not structure: they visualise the depth degeneracy of Sec. 6 — opacity that the two-view geometry cannot localise in height, smeared along the viewing directions.

The uncertainty is concentrated in one axis. Free to place material anywhere along the two viewing cones, the solver spreads it in height: only **28%** of the interior opacity mass lies within 6–8 m, the vertical profile peaks near **4.2 m** rather than 7.0 m, and the beam amplitude at 7.0 m is $0.012\,\mathrm{m}^{-1}$ against $0.187\,\mathrm{m}^{-1}$ in the layered solve (~15×). This is a *known, quantified and correctable* geometric limit: pinning the solve at the autofocused height restores the full amplitude (Sec. 6).

quantity	layered (7 m)	full-3D
beam amplitude at $z = 7$ m (m^{-1})	0.187	0.012
beam / inter-beam contrast	6.9	5.0
interior mass in $z = 6$–8 m	≈ 100%	28%
vertical mass-profile peak (m)	7.0	4.2
beam-position offset vs data (m)	0.10	— (faint)

In the interactive viewer the model is rendered as a density-coloured voxel cloud with adjustable cutoff and height cut (Fig. 4), so every claim above can be inspected directly.

6 Uncertainties, and what more detectors would buy

(i) Depth degeneracy. Height information comes solely from the parallax between two viewpoints. With baseline $b \approx 1.9$ m and lever arm $z \approx 7$ m, a feature's inter-view displacement changes by only $b/z \approx 0.27$ of its lateral extent per unit height error, so the angular bin width translates into a height uncertainty $\delta z \approx z^2\sigma_\theta/b$ of order a metre. An unconstrained solve fills this null space with smooth vertical smears — the veils of Fig. 4 — and cannot state *where* in height the material sits, only along *which sightlines* it lies.

(ii) Periodic-pattern aliasing. As quantified in Sec. 3.5, a repeating pattern admits false height solutions spaced by pitch × z/b. Fortuitously distant here (≈ 5.8 m), it is a design parameter that must be checked for any new geometry; a shorter pitch or baseline can place the alias inside the plausible range, where only physics-based cross-validation resists it.

(iii) Boundary artifacts. Rim voxels constrained by few grazing rays accumulate spurious mass up to an order of magnitude above the interior. Untreated, the ring dominates display normalisation and renders the genuine beams invisible (an “inverted” appearance); it must be masked or tapered before interpretation.

(iv) No redundancy. With two views there is no third to arbitrate: cross-position validation is the *only* internal consistency check, and a single detector alone is entirely height-blind, so pose or calibration errors project into the volume unexposed.

The contribution of additional detectors. All four uncertainties are geometric, so they respond to viewpoints, not exposure — doubling the statistics with two views sharpens the lateral picture but recovers no depth. The scaling $\delta z \approx z^2\sigma_\theta/b$ makes the return concrete: a **third detector** on a ~5 m baseline cuts the depth uncertainty from ~1 m to ~0.4 m, comparable to the beam depth itself, and placed non-collinearly it provides parallax in both lateral directions at once. A **four-corner array** ($b \approx 8$–10 m) reaches $\delta z \approx 0.2$ m — true sharp voxels — while k viewpoints supply $k(k-1)/2$ baselines, so redundancy grows faster than hardware: independent consistency checks, jointly self-calibrating poses, and rim voxels crossed by multiple ray fans, suppressing the boundary ring at the source. Even *repeat runs of the same two detectors at displaced positions* buy most of this at zero hardware cost.

7 Conclusion

Two floor-mounted muon detectors — a minimal stereo pair — suffice to build a complete-room voxel model that gets the ceiling right where the data constrain it: every beam at its model-free verified position (0.10 m), correct material/air segmentation over 1.9 M voxels, and a structure height measured from the data itself and validated on synthetic ground truth to ~0.2 m. The residual uncertainty lives in a single, well-understood axis — depth — where the short baseline smears the vertical mass profile (28% within 2 m of the true ceiling); the layered solve at the autofocused height corrects it by construction. The path forward is concrete: one added viewpoint shrinks the depth uncertainty ~3×, a four-corner array reaches beam-scale sharpness. That a result this close to the truth comes from two boxes of muon shadows is the strongest argument for the method — and for the third detector.

References

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Artifacts: `runs/production/{volume_full3d.npz, volume.npz, viewer.html, autofocus_report.pdf}`, `reports/autofocus_validation.png`. Reproduce: `python -m muontomo.reconstruct --config configs/full3d.json`; autofocus validation: `python scripts/autofocus_validation.py`.